WEEK 7, BEFORE THE FIRST PASS

Week 7 pre-read.

About 30 minutes. Graded complete or incomplete. Do this before the first pass.

This is a quick look ahead, not a full read. Look at the headings and the figures, answer the questions below in a few words, and move on. You will learn all of it properly in the walkthroughs, and it will make much more sense because you have already seen it once. **This week covers parts of three chapters: Chapter 13 (muscles), Chapter 12 (the motor neuron and the neuromuscular junction), and the rest of Chapter 14 (how the brain commands a movement).**

Write every answer in your own words

Do not copy from the book, the notes, a website, a classmate or an AI tool, even if you change a few words. Read the source, look away, and then write what you understood. If you cannot say it without looking, that tells you what to go back and study. Your own words, even rough ones, are what I want to see.

Where to look

Read closely, Silverthorn Chapter 13, Muscles: the whole section on skeletal muscle, and the section on the mechanics of body movement. That covers how a muscle is built, how a nerve signal makes it contract, where its energy comes from, fatigue, fiber types, twitches and motor units. The sections on smooth muscle and cardiac muscle come later in the course, with the heart, so skip them for now.

Read, Silverthorn Chapter 12, Efferent Division: Autonomic and Somatic Motor Control: only the section on the somatic motor division, which covers the motor neuron and the neuromuscular junction. The autonomic part of this chapter is next week.

Read, Silverthorn Chapter 14, Integrative Physiology I: Control of Body Movement: the section on the integrated control of body movement, which comes after the reflexes you read last week. It covers how the brain plans, starts and adjusts a movement. Skip the last section, on movement in visceral muscles.

Chapter numbers are for the 9th edition. Other editions number things differently, so go by the chapter title and the topic, and use the search in your eText to find it.

No Silverthorn textbook? Use OpenStax

If you chose not to buy the textbook, use the free OpenStax book, Anatomy and Physiology 2e, and do the same pre-read with it. Read these sections:

- **10.2 Skeletal Muscle.** How a skeletal muscle is built, from the whole muscle down to the sarcomere.
- **10.3 Muscle Fiber Contraction and Relaxation.** How the filaments slide, what calcium does, and where the energy comes from.
- **10.4 Nervous System Control of Muscle Tension.** Twitches, summation, tetanus, motor units and the length of the muscle.
- **10.5 Types of Muscle Fibers.** Slow and fast fibers and what each one is good for.
- **10.6 Exercise and Muscle Performance.** What training and disuse do to a muscle.
- **14.3 Motor Responses.** How the brain and spinal cord command and adjust a movement.

About the figures. The figures listed on this page are the ones in Silverthorn. For each one, look for a picture of the same thing in those OpenStax sections. If you cannot find a matching picture, write your own list instead: on a sheet of paper, list each picture you did find in those sections and write one sentence about what it shows. Upload that page with your pre-read.

The Find it questions still work with OpenStax. Search the OpenStax sections for the same key words.

Figures to look at closely

These are the most important figures for this week. Find each one in your eText by what it shows, using the search for the words in bold. Read its labels, and look for the thing named after it.

Chapter 13, skeletal muscle

- **The levels of a skeletal muscle.** The figure that goes from the whole muscle to a bundle of fibers, one fiber, a myofibril and the thick and thin filaments. Find the T tubules and the sarcoplasmic reticulum around each myofibril.
- **The sarcomere.** Find the Z disc at each end, the dark A band where the thick filaments are, and the light I band that has only thin filaments.
- **The thin filament.** Find actin, the long tropomyosin strand that lies along it, and troponin, the small complex that holds tropomyosin in place.
- **Excitation–contraction coupling.** Follow the action potential down the T tubule. Find the two receptors that touch each other there, one in the T tubule and one in the sarcoplasmic reticulum.
- **The crossbridge cycle.** Find the step where ATP binds to the myosin head and the step where ATP is split. They are not the same step.
- **The muscle twitch.** Line up the muscle action potential with the tension it makes. Find the latent period.
- **Summation and tetanus.** Compare a single twitch, unfused tetanus and fused tetanus.
- **The length–tension relationship.** Find the sarcomere length that gives the most tension.

Chapter 12, the neuromuscular junction

- **The neuromuscular junction.** Number the events in order, from the action potential arriving at the axon terminal to the muscle fiber firing its own action potential.

Chapter 14, control of movement

- **The integration of movement.** Find what the cerebellum and the basal ganglia each add, and where the command leaves the brain for the spinal cord.

What this reading should help you do

Read each goal and pick the answer that fits you right now. There is no wrong answer here.

Muscle and the neuromuscular junction (Chapters 13 and 12)

Trace a signal from a motor neuron across the neuromuscular junction to an action potential in the muscle fiber.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Explain how that action potential releases calcium inside the fiber, and how calcium switches on contraction.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Put the steps of the crossbridge cycle in order and say where ATP is used.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Explain what ends a contraction and lets the muscle relax.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Explain how a muscle makes a small force or a large one: twitches, summation and motor units.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Compare the three kinds of muscle fiber and the three ways a muscle makes ATP, and explain fatigue.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Control of movement (Chapter 14)

Describe how the brain plans, starts and adjusts a movement, and what the cerebellum and basal ganglia each do.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Tell an upper motor neuron problem from a lower motor neuron problem by tone, reflexes and muscle size.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Find it

Each answer is in the part of the chapter named. A few words is enough.

Muscle and the neuromuscular junction (Chapters 13 and 12)

What is a motor unit? Why do the muscles that move your eyes have small motor units?

In Chapter 13, the part on motor units. Search your eText for motor unit.

At the neuromuscular junction, what neurotransmitter is released, what kind of receptor does it bind, and which way do sodium ions move?

In Chapter 12, the section on the somatic motor division. Search for neuromuscular junction.

Name the receptor in the T tubule and the receptor in the sarcoplasmic reticulum that work together to release calcium.

In Chapter 13, the part on excitation-contraction coupling. Search for ryanodine.

What does calcium bind to, and what does that binding move out of the way?

In Chapter 13, the part on how calcium starts contraction. Search for troponin.

In the crossbridge cycle, what does the binding of ATP do to the myosin head? What does splitting ATP do?

In Chapter 13, the part on the crossbridge cycle. Search for power stroke.

What is rigor mortis, and why does it happen?

Same part of Chapter 13. Search for rigor.

What is the latent period of a muscle twitch, and what is happening inside the fiber during it?

In Chapter 13, the part on the muscle twitch. Search for latent period.

What is the difference between unfused (incomplete) tetanus and fused (complete) tetanus?

In Chapter 13, the part on summation. Search for tetanus.

Name the three kinds of skeletal muscle fiber. Which one resists fatigue best, and why?

In Chapter 13, the part on fiber types. Look for the table that compares them.

What is the difference between an isotonic and an isometric contraction? Give an example of each.

In Chapter 13, the section on the mechanics of body movement. Search for isometric.

Control of movement (Chapter 14)

What does the cerebellum add to a movement, and what do the basal ganglia add?

In Chapter 14, the section on the integrated control of body movement. Search for cerebellum.

Words to know

Find each term in the reading. Write what it means in your own words, not the book's. If you are working on paper, add a quick sketch. If you are typing, give one example instead.

Sarcomere

Motor unit

End plate potential

Excitation–contraction coupling

Crossbridge

Tetanus

Recruitment

Creatine phosphate

Muscle fatigue

Corticospinal tract

Before you start the walkthroughs

Which of this week's topics looks hardest to you right now, and why?

Write one question you want the walkthroughs to answer.

Turn it in

Save this page as a PDF with the button below and upload it to the Week 7 pre-read assignment in Canvas. If you worked on paper, upload a photo of your pages instead.